

# Simple Machines — Workout

ICSE Class 6 Physics · 10 MCQs · 1 mark each · Name: \_\_\_\_\_ Date: \_\_\_\_\_

**Syllabus:** Whole chapter: work & machines, MA, VR & efficiency, levers and their orders, pulley, wheel and axle, inclined plane, wedge and screw.

**Instructions:** Each question has one correct option. Tick or circle the letter of the best answer. Attempt all questions.

**1. A porter carrying a box on his head walks across a flat platform. The work he does against gravity is:**

- |            |                              |
|------------|------------------------------|
| A. maximum | B. equal to the box's weight |
| C. zero    | D. negative                  |

**2. Which of these is NOT a unit of energy?**

- |         |                 |
|---------|-----------------|
| A. watt | B. newton metre |
| C. erg  | D. joule        |

**3. A machine lifts a 720 N load with a 180 N effort. Its mechanical advantage is:**

- |        |         |
|--------|---------|
| A. 4   | B. 900  |
| C. 540 | D. 0.25 |

**4. A machine has MA = 3 and VR = 4. Its efficiency is:**

- |        |         |
|--------|---------|
| A. 7%  | B. 133% |
| C. 12% | D. 75%  |

**5. Which of these is a first-order lever?**

- |                  |                              |
|------------------|------------------------------|
| A. a wheelbarrow | B. a crowbar lifting a stone |
| C. a nutcracker  | D. sugar tongs               |

**6. A wheelbarrow is a second-order lever, so its mechanical advantage is always:**

- |                |                   |
|----------------|-------------------|
| A. zero        | B. greater than 1 |
| C. less than 1 | D. equal to 1     |

**7. Which of these is a third-order lever?**

- |                    |                       |
|--------------------|-----------------------|
| A. a fishing rod   | B. a see-saw          |
| C. a bottle opener | D. a pair of scissors |

**8. The ideal mechanical advantage of a single movable pulley is:**

- |        |      |
|--------|------|
| A. 0.5 | B. 2 |
| C. 1   | D. 4 |

**9. A 600 N drum is rolled up a 3 m plank to a height of 1 m. The ideal effort needed is:**

- |          |           |
|----------|-----------|
| A. 600 N | B. 1800 N |
| C. 100 N | D. 200 N  |

**10. A nail being driven into wood and an axe splitting a log both act as:**

- |           |                     |
|-----------|---------------------|
| A. levers | B. a wheel and axle |
| C. wedges | D. pulleys          |

## Answer Key

1. C

2. A

3. A

4. D

5. B

6. B

7. A

8. B

9. D

10. C